

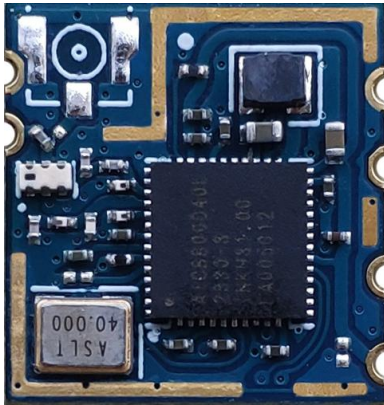


BL-M8800DU6-40L

**802.11ax 287Mbps WLAN + BLE v5.4
USB Module Specification**

SHENZHEN BILIAN ELECTRONIC CO., LTD

Add: 10~11/F, Building 1A, Huaqiang idea park, Guangming district, Shenzhen. Guangdong, China
Web: www.b-link.net.cn



(Top View)



(Bottom View)

Module Name: BL-M8800DU6-40L

Module Type: 802.11a/b/g/n/ac/ax 287Mbps WLAN +BLE v5.4 USB Module

Revision: V1.1.2

Customer Approval:

Company:

Title:

Signature:

Date:

Approval:

Title:

Signature:

Date:

Revision History

Revision	Summary	Release Date	Revised By
V0.1.0	Initial release	2024-01-23	Xhg
V1.0.0	Official release	2024-03-25	Xhg
V1.1.1	Update the product appearance、Pin Assignment and Mechanical Dimensions	2025-04-28	Cxf
V1.1.2	1、Add Product appearance description 2、Update Product images	2025-09-18	Wlz

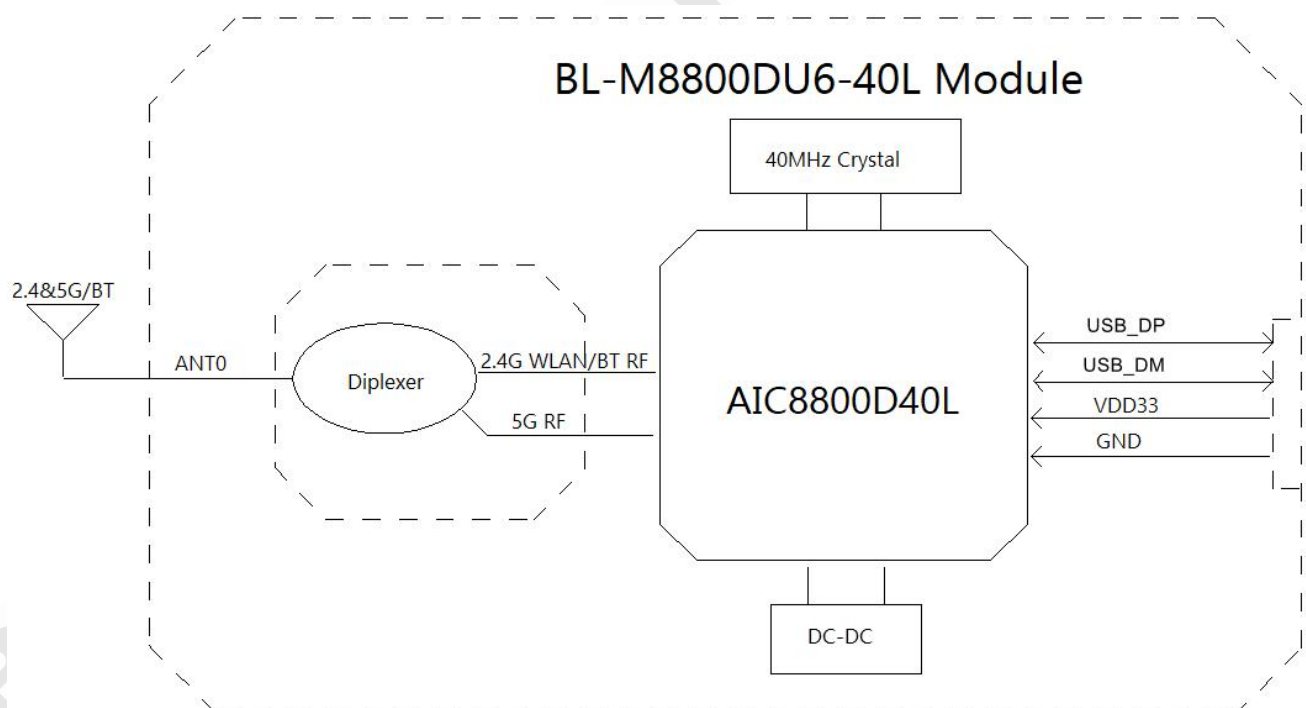
1. Introduction

BL-M8800DU6-40L is a highly integrated Dual-band WLAN+Bluetooth Combo module. It combines a 1T1R Dual-band WLAN subsystem and a BLE v5.4 subsystem. This module compatible IEEE 802.11 a/b/g/n/ac/ax standard and provides the maximum PHY rate up to 287Mbps, offering feature-rich wireless connectivity at high standards, and delivering reliable, cost-effective, high throughput from an extended distance.

1.1 Features

- Operating Frequencies: 2.4~2.4835GHz or 5.15~5.85GHz
- Wireless PHY rate can reach up to 287Mbps with 20/40MHz bandwidth
- Host Interface is USB2.0
- Support STA, AP, WLAN Direct modes concurrently
- Support BT Low Energy mode only

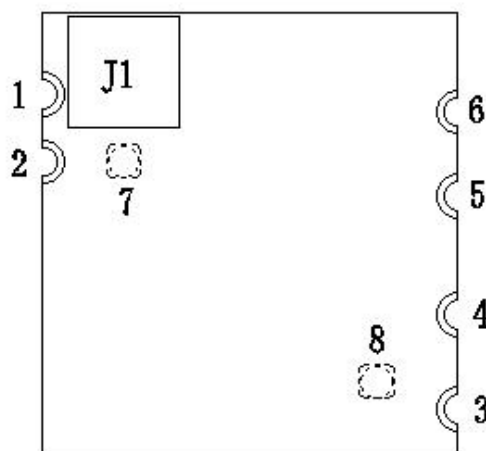
1.2 Block Diagram



1.3 General Specifications

Module Name	BL-M8800DU6-40L
Chipset	AIC8800D40L
WLAN Standards	IEEE802.11a/b/g/n/ac/ax
Host Interface	USB2.0 for WLAN & Bluetooth
Antenna	Connect to the external antenna through half hole pad
Dimension	13.0*12.3*1.8mm (L*W*H)
Power Supply	3.3V±0.2V main power supply @700mA (Max)
Operation Temperature	-20°C to +70°C
Operation Humidity	10% to 95% RH (Non-Condensing)

2. Pin Assignments



(Top view)

2.1 Pin Definition

No	Pin Name	Type	I/O Level	Module Pin Description
1	GND	RF	/	RF Ground connections
2	ANT0	RF	/	RF Pad for 2.4G WLAN/5G WLAN/2.4G BT ANT
3	VDD33	P	/	3.3V main power supply
4	USB_DM	A I/O	/	USB 2.0 Device High Speed Interface differential pair

5	USB_DP	A I/O	/	USB 2.0 Device High Speed Interface differential pair
6	GND	P	/	Ground connection
7	GND	RF	/	Auxiliary RF Ground connection PAD for ANT (It can be left floating, but grounding is strongly recommended to enhance ground connectivity and avoid RF interference)
8	GND	P	/	Auxiliary Ground connection PAD for power and USB (It can be left floating, but grounding is strongly recommended to reduce the power loop area and avoid interference with RF)
	J1	RF	/	NC(Reserved IPEX for 2.4G WLAN/5G WLAN/2.4G BT to ANT)

P: Power or Ground; I: Input; O :Output; A I/O: Analog In/Output; RF: Analog RF Port or RF Ground

3. Electrical and Thermal Specifications

3.1 Recommended Operating Conditions

Parameters		Min	Typ	Max	Units
Ambient Operating Temperature		-20	25	70	°C
External Antenna VSWR			1.7	2.1	/
Supply Voltage	VDD33	3.1	3.3	3.5	V

3.2 Current Consumption

Conditions : VDD33=3.3V; Ta:25°C;			
Use Case	VDD33 Current		
	Typ(I _{RMS})	Max(I _{Peak})	Units
WLAN Unassociated (Linux USB Driver)	85	96	mA
2.4G WLAN throughput 180Mbps (Linux USB Driver)	186	350	mA
5G WLAN throughput 180Mbps (Linux USB Driver)	244	550	mA
2.4G 11b 1Mbps TX@18dBm (TX RF test)	96	309	mA
2.4G 11g 6Mbps TX@18dBm (TX RF test)	120	349	mA
2.4G 11n HT20_MCS0 TX@17dBm (TX RF test)	102	337	mA
2.4G 11ax HE_SU 40M_MCS0 TX@15dBm (TX RF test)	84	306	mA

2.4G 11ax HE_SU 40M_MCS11 TX@13dBm (TX RF test)	67	285	mA
2.4G 11ax HE_SU 40M_MCS11 RX (RX RF test)	76	92	mA
5G 11a 6Mbps TX@17dBm (TX RF test)	171	528	mA
5G 11n HT20_MCS0 TX@17dBm (TX RF test)	160	538	mA
5G 11ax HE_SU 40M_MCS11 TX@13dBm (TX RF test)	87	388	mA
5G 11ax HE_SU 40M_MCS11 RX (RX RF test)	86	104	mA
BT LE_1M TX@7dBm (BT RF test, WLAN disable)	45	63	mA
BT LE_1M RX Active (BT RF test, WLAN disable)	32	47	mA

4. WLAN & Bluetooth RF Specifications

4.1 2.4G RF Specification

Conditions : VDD33=3.3V ; Ta:25°C			
Features		Description	
WLAN Standard		IEEE 802.11b/g/n/ax	
Frequency Range		2.4~2.4835GHz (2.4GHz ISM Band)	
Channels		Ch1~Ch13 (For 20MHz Channels)	
Modulation		802.11b (DSSS): DBPSK, DQPSK, CCK; 802.11g (OFDM): BPSK, QPSK, 16QAM, 64QAM; 802.11n (OFDM): BPSK, QPSK, 16QAM, 64QAM; 802.11ax (OFDMA): BPSK, BPSK_DCM, QPSK, QPSK_DCM, QAM16, QAM16_DCM, QAM64, QAM256, QAM1024;	
Data Rate		802.11b: 1, 2, 5.5, 11Mbps; 802.11g: 6, 9, 12, 18, 24, 36, 48, 54Mbps; 802.11n (HT20): MCS0~MCS7(1T1R_SISO) 6.5~72.2Mbps; 802.11n (HT40): MCS0~MCS7(1T1R_SISO) 13.5~150Mbps; 802.11ax (HE_MU,OFDMA 26~242RU): MCS0~MCS11(1T1R) 0.4~143.4Mbps; 802.11ax (HE_SU, non-OFDMA 20MHz): MCS0~MCS11(1T1R) 3.6~143.4Mbps; 802.11ax (HE_SU, non-OFDMA 40MHz): MCS0~MCS11(1T1R) 7.3~286.8Mbps;	
Frequency Tolerance		≤ ±20ppm	
2.4G Transmitter Specifications (TX power tolerance calibrated, customers can define the target TX power within recommended range by modifying configuration file of the driver software, Customers must define the TX power same or lower than recommended Target TX Power as below.)			
TX Rate	Recommended Target TX Power (dBm)	TX Power Tolerance (dBm)	EVM (dB)

802.11b@1Mbps	18	±2	≤-10
802.11b@11Mbps	18	±2	≤-10
802.11g@6Mbps	18	±2	≤-10
802.11g@54Mbps	16	±2	≤-25
802.11n@HT20_MCS0	17	±2	≤-10
802.11n@HT20_MCS7	15	±2	≤-28
802.11n@HT40_MCS0	17	±2	≤-10
802.11n@HT40_MCS7	15	±2	≤-28
802.11ax@HE_SU 40M_MCS0	17	±2	≤-15
802.11ax@HE_SU 40M_MCS11	13	±2	≤-35

2.4G Receiver Specifications

RX Rate	Min Input Level (Typ. dBm)	Max Input Level (Typ. dBm)	PER
802.11b@1Mbps	-96	-5	< 8%
802.11b@11Mbps	-88	-5	< 8%
802.11g@6Mbps	-92	-5	< 10%
802.11g@54Mbps	-75	-5	< 10%
802.11n@HT20_MCS0	-92	-5	< 10%
802.11n@HT20_MCS7	-73	-5	< 10%
802.11n@HT40_MCS0	-89	-5	< 10%
802.11n@HT40_MCS7	-71	-5	< 10%
802.11ax@HE_SU 40M_MCS0	-88	-5	< 10%
802.11ax@HE_SU 40M_MCS11	-60	-5	< 10%

4.2 5G WLAN RF Specification

Conditions: VDD33=3.3V; Ta:25°C	
Features	Description
WLAN Standard	IEEE 802.11a/n/ac/ax
Frequency Range	5.15~5.25GHz; 5.25~5.35GHz; 5.47~5.725GHz; 5.725~5.85GHz (5GHz ISM Band)

Channels	Ch36, Ch40, Ch44, Ch48; Ch52~Ch64; Ch100~Ch140; Ch149~Ch165(For 20MHz Channels)		
Modulation	802.11a (OFDM): BPSK, QPSK, QAM16, QAM64; 802.11n (OFDM): BPSK, QPSK, QAM16, QAM64; 802.11ac (OFDM): BPSK, QPSK, QAM16, QAM64, QAM256; 802.11ax (OFDMA): BPSK, BPSK_DCM, QPSK, QPSK_DCM, QAM16, QAM16_DCM, QAM64, QAM256, QAM1024;		
Data Rate	802.11a: 6, 9, 12, 18, 24, 36, 48, 54Mbps; 802.11n (HT20): MCS0~MCS7 6.5~72.2Mbps; 802.11n (HT40): MCS0~MCS7 13.5~150Mbps; 802.11ac (VHT20): MCS0~MCS8(1T1R) 6.5~86.7Mbps; 802.11ac (VHT40): MCS0~MCS9(1T1R) 13.5~200Mbps; 802.11ax (HE_MU, OFDMA 26~242RU): MCS0~MCS11(1T1R) 0.4~143.4Mbps; 802.11ax (HE_SU, non-OFDMA 20MHz): MCS0~MCS11(1T1R) 3.6~143.4Mbps; 802.11ax (HE_SU, non-OFDMA 40MHz): MCS0~MCS11(1T1R) 7.3~286.8Mbps;		
Frequency Tolerance	≤ ±20ppm		
5G Transmitter Specifications (TX power tolerance calibrated, customers can define the target TX power within recommended range by modifying configuration file of the driver software, Customers must define the TX power same or lower than recommended Target TX Power as below.)			
TX Rate	Recommended Target TX Power (dBm)	TX Power Tolerance (dBm)	EVM (dB)
802.11a@6Mbps	17	±2	≤ -10
802.11a@54Mbps	16	±2	≤ -25
802.11n@HT20_MCS0	17	±2	≤ -10
802.11n@HT20_MCS7	15	±2	≤ -28
802.11n@HT40_MCS0	17	±2	≤ -10
802.11n@HT40_MCS7	15	±2	≤ -28
802.11ac@VHT_40M_MCS0	16	±2	≤ -10
802.11ac@VHT_40M_MCS9	14	±2	≤ -32
802.11ax@HE_SU 40M_MCS0	16	±2	≤ -15
802.11ax@HE_SU 40M_MCS11	13	±2	≤ -35
5G Receiver Specifications			

RX Rate	Min Input Level (Typ. dBm)	Max Input Level (Typ. dBm)	PER
802.11a@6Mbps	-90	-5	< 10%
802.11a@54Mbps	-74	-5	< 10%
802.11n@HT20_MCS0	-90	-5	< 10%
802.11n@HT20_MCS7	-71	-5	< 10%
802.11n@HT40_MCS0	-88	-5	< 10%
802.11n@HT40_MCS7	-68	-5	< 10%
802.11ac@VHT_40M_MCS0	-87	-5	< 10%
802.11ac@VHT_40M_MCS9	-62	-5	< 10%
802.11ax@HE_SU 40M_MCS0	-86	-5	< 10%
802.11ax@HE_SU 40M_MCS11	-59	-5	< 10%

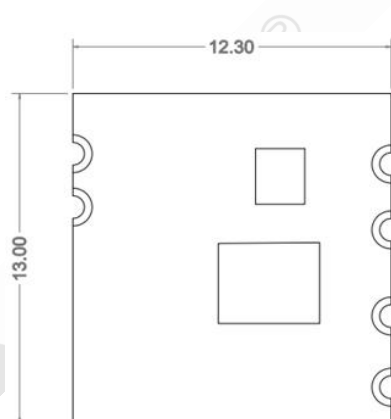
4.3 Bluetooth RF Specification

Conditions: VDD33 =3.3V; Ta:25°C			
Features	Description		
Bluetooth Specification	Bluetooth Core Specification v5.4/v4.2/v4.0		
Frequency Range	2.4~2.4835GHz(2.4GHz ISM Band)		
Channels	Blue Low Energy: Ch0~Ch39 (For 2MHz Channels);		
Power Classes	Bluetooth Low Energy: Class1.5;		
Data Rate & Modulation	LE_125Kbps: GFSK (Coded_S=8); LE_500Kbps: GFSK (Coded_S=2); LE_1Mbps: GFSK (Uncoded); LE_2Mbps: GFSK (Uncoded);		
Bluetooth Transmitter Specifications			
Items	Min (dBm)	Typ (dBm)	Max (dBm)
TX Power			
LE_125/500K	3	7	10
LE_1M	3	7	10
LE_2M	3	7	10
LE_1M Modulation Characteristics			

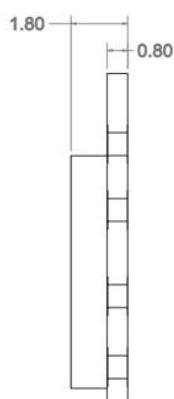
Δf1avg	225KHz	248.75KHz	275KHz	
Δf2avg	185KHz	224.33KHz	275KHz	
Δf2max	185KHz	202.84KHz	/	
Δf2avg/Δf1avg	0.8	0.90	/	
LE_2M Modulation Characteristics				
Δf1avg	450KHz	499.16KHz	550KHz	
Δf2avg	370KHz	472.22KHz	550KHz	
Δf2max	370KHz	441.75KHz	/	
Δf2avg/Δf1avg	0.8	0.94	/	
Bluetooth Receiver Specifications				
Items	Sensitivity		Maximum Input Level	
	Input Level (Typ. dBm)	PER	Input Level (Typ. dBm)	PER
LE_125/500K	-90	≦ 5%	-5	≦ 5%
LE_1M	-88	≦ 5%	-5	≦ 5%
LE_2M	-84	≦ 5%	-5	≦ 5%

5. Mechanical Specifications

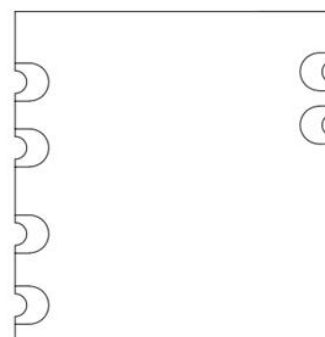
5.1 Module Outline Drawing



(Top View)

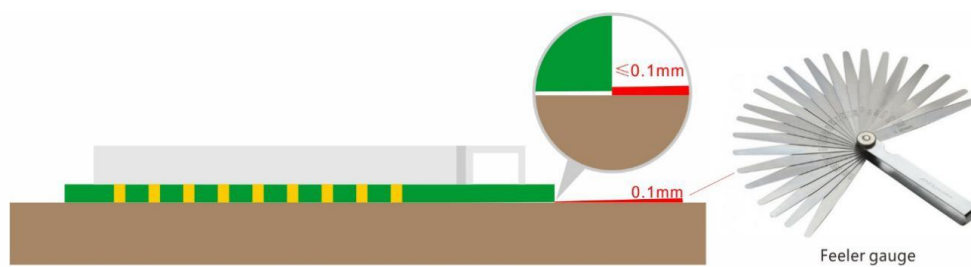


(Side View)



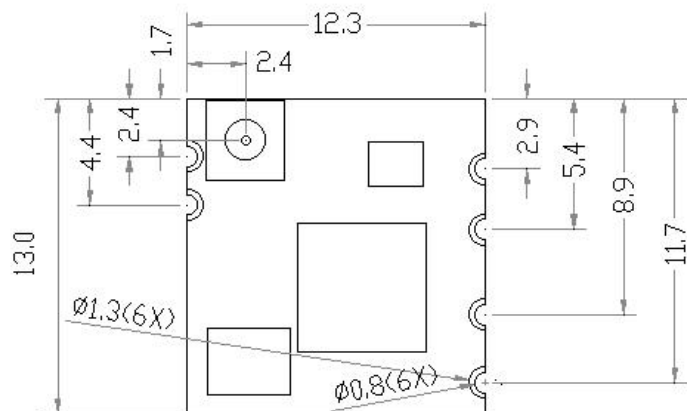
(Bottom View)

Module dimension: 13.0*12.3*1.8mm (L*W*H; Tolerance: $\pm 0.3\text{mm}_L/W$, $\pm 0.2\text{mm}_H$)

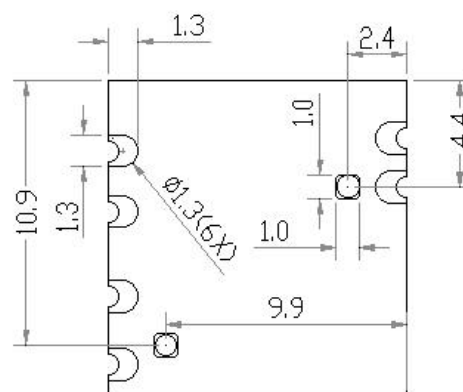


Module Bow and Twist: $\leq 0.1\text{mm}$

5.2 Mechanical Dimensions



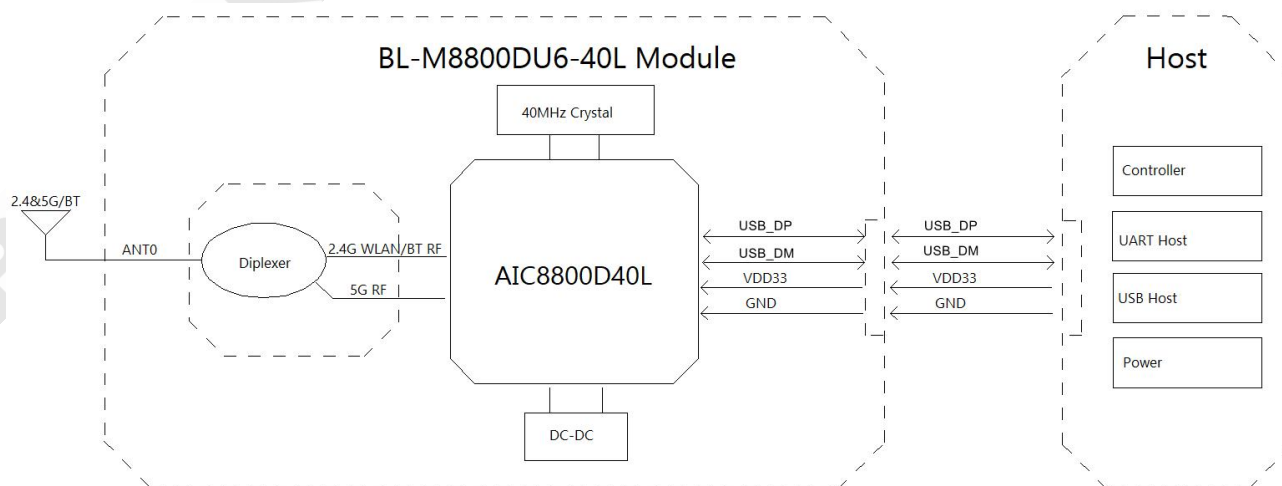
(Top View)



(Bottom View)

6. Application Information

6.1 Typical Application Circuit



6.2 HW Application Note

The module uses general packaging of USB_13.0*12.3mm, which has the defect that GND Pins are too far away from RF/USB/Power Pins. Therefore, in the application of RF from half hole PAD, the power loop may interfere with the RF loop and cause RF performance derating, customers must be careful to design RF, USB and Power circuits of the module on their motherboard!

6.2.1 VDD33 Power requirement:

- A、DC 3.1~3.5V & Ripple Voltage <100mV power supply input, Maximum RMS current $\geq$ 500mA and Maximum Peak current $\geq$ 700mA.
- B、For achieve fast transient response, a current mode buck converter recommended.
- C、On customer's motherboard, use 10uF and 100nF MLCC capacitors close to the module's VDD33 Pin for power input decoupling. Each GND Pin has three close vias to ensure connectivity and thermal conductivity.

6.2.2 USB interface Design Guidelines:

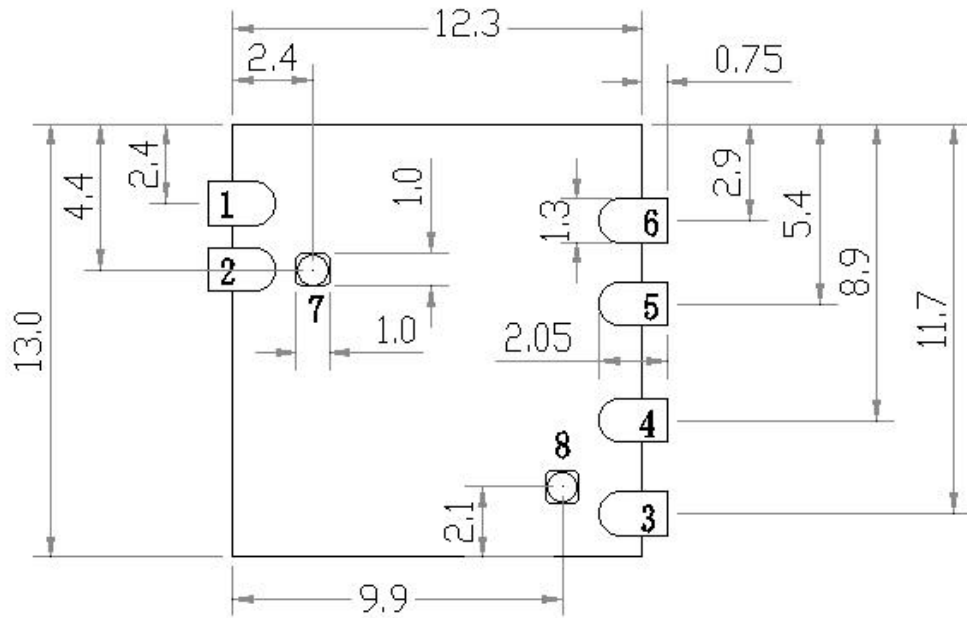
- A、The module provides a USB device interface which is compatible with USB2.0 specification, High-Speed mode supports data transmission rate up to 480Mbps.
- B、On customer's motherboard, PCB traces of the USB high-speed signal pair should be maintain 90 Ω differential impedance, structure of "Differential Coated Coplanar Waveguide With Ground" with the advantages of impedance control and GND surrounding isolation interference may be an ideal choice. To avoid interference, USB signal pair must be far away from power, RF and other signals, GND copper can be used to surround and isolate them.
- C、PCB traces of the USB high-speed signal pair as short as possible, as far away from other signal as possible, minimize the length mismatch of signal pair, avoid layer change and maintain a complete reference layer to reduces signal reflections and impedance changes.
- D、If It is necessary to change layers for USB signal pair routing, use GND vias close to signal pair's vias as the shortest return path.

6.2.3 RF Path Design Guidelines:

- A、Soldering two auxiliary grounding PADs at the bottom of module to customer's motherboard can significantly improve RF performance derating especially the poor RX sensitivity.
- B、On customer's motherboard, keep complete GND copper area under and around the module, do not route Power, RF, USB and other signals there to avoid interfering with the module causing RF performance derating problem.
- C、PCB traces of the RF path as short as possible, as far away from other signal as possible, avoid layer change, and should be maintain continuous 50 Ω characteristic impedance, structure of "Coated Coplanar Waveguide With Ground" with the advantages of impedance control and GND surrounding isolation interference may be an ideal choice. It is necessary to maintain a complete GND reference layer below the RF path area.
- D、RF Pin and antenna PAD/RF connector PAD on customer's motherboard are significantly wider than PCB traces of the RF path, which will cause RF impedance discontinuity. To improve this problem, in addition to

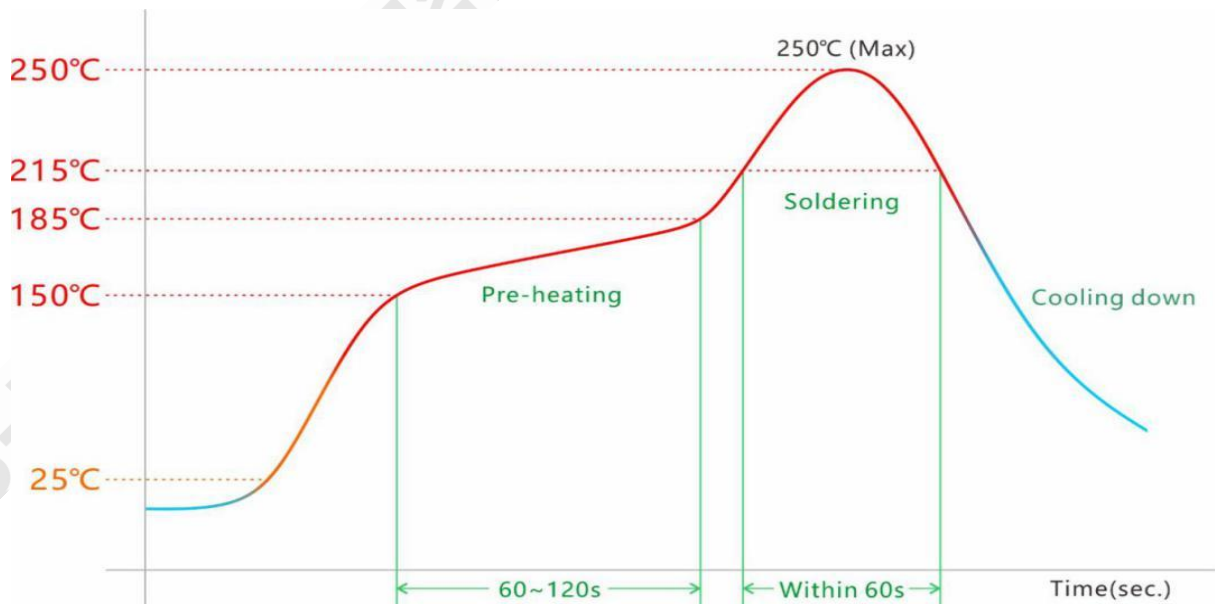
adjusting the local impedance of these PADs, a “ π ” circuit can be reserved near the antenna/RF connector which can match both RF path impedance and antenna impedance.

6.3 Recommend PCB Layout Footprint



(Design Unit: mm)

6.4 Reflow Soldering Standard Conditions



Please use the reflow within 2 times.
Set up the highest temperature within 250°C.

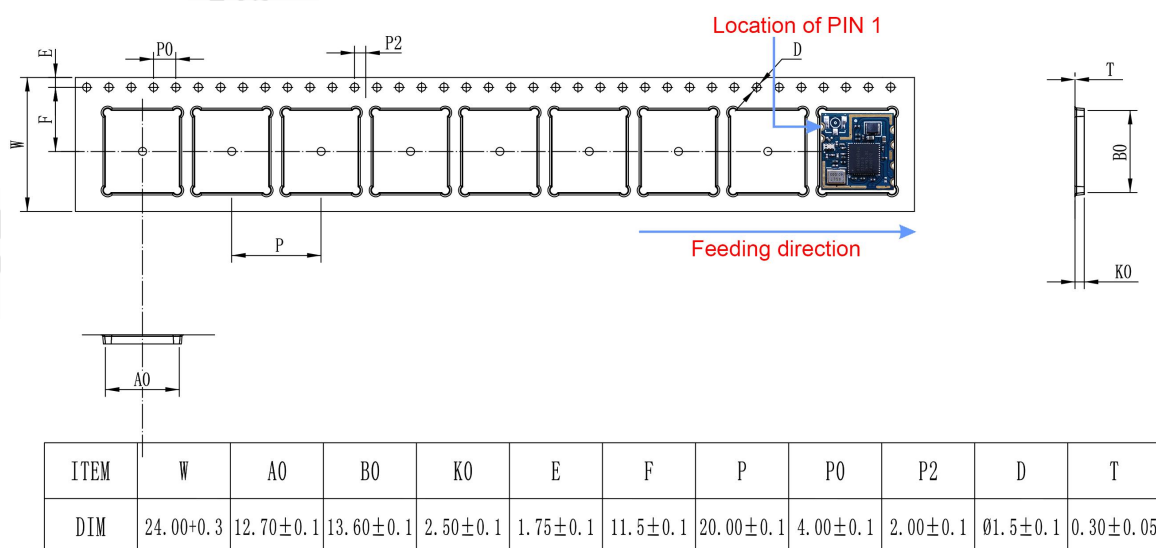
7. Key Components Of Module

No.	Parts	Specification	Manufacturer	Note
1	Chipset	AIC8800D40L	AIC Semiconductor (Shanghai) CO.,LTD	
2	PCB	BL-M8800DU6	SHEN ZHEN QILI ELECTRON CO.,LTD	
			Quzhou Sunlord Electronics Co.,Ltd	
			ShenZhen Tie Fa Technology Limited	
3	Crystal	40MHz-3225	LUCKI CM ELECTRONICS CO.,LTD	
			Chengde oscillator Electronic Technology CO.,LTD	
			JinHua East Crystal Electronic CO.,LTD	
			HUBEI TKD CRYSTAL ELECTRONIC SCIENCE AND TECHNOLOGY CO.,LTD	
4	Diplexer	DIP1608	Shenzhen Sunlord Electronics Co.,Ltd	
			Dongguan Hekang Electronics Co.,LTD	
			Shenzhen FTR Technologies CO., LTD	
			Walsin Technology Corporation	

Note: To ensure supply chain stability, certified alternative components with potential cosmetic variations have been utilized. The physical appearance of the module shall be governed by the final delivered unit.

8. Package and Storage Information

8.1 Package Dimensions





Package specification:

1. 1,000 modules per roll and 5,000 modules per box.
2. Outer box size: 37.5*36*29cm.
3. The diameter of the blue environment-friendly rubber plate is 13 inches, with a total thickness of 28mm (with a width of 24mm carrying belt).
4. Put 1 package of dry agent (20g) and 1 humidity card in each anti-static vacuum bag.
5. Each carton is packed with 5 boxes.

8.2 Storage Conditions

Absolute Maximum Ratings:

Storage temperature: -40°C to +85°C

Storage humidity: 10% to 95% RH (Non-Condensing)

Recommended Storage Conditions:

Storage temperature: 5°C to +40°C

Storage humidity: 20% to 90% RH

Please use this Module within 12month after vacuum-packaged.

The Module shall be stored without opening the packing.

After the packing opened, the Module shall be used within 72hours.

When the color of the humidity indicator in the packing changed, the Module shall be baked before soldering.

Baking condition: 60°C, 24hours, 1time.

ESD Sensitivity:

ESD Protection: 2KV(HBM, Maximum rating)

The Module is a static-sensitive electronic device.

Do not operate or store near strong electrostatic fields.

Take proper ESD precautions!



ESD CAUTION